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Classification of AI Task Environments

1 Introduction

An AI task environment can be classified according to its observability and the predictability of its outcomes. **Observability** describes how much information an agent can obtain about the current state of its environment. **Determinism** describes whether an action produces a predictable outcome or whether uncertainty may affect the result.

The two systems specified in Question 1—a delivery robot and an LLM-based student-support agent—are classified below.

2 Delivery Robot

A. Observability: Partially Observable

The delivery robot operates in a partially observable environment because its sensors cannot provide complete information about every aspect of its surroundings.

- **Observable variable:** The distance to a nearby obstacle detected by a LiDAR sensor.
- **Hidden variable:** A pedestrian approaching behind a closed door or around a corner.

Although cameras and distance sensors provide useful information, obstacles may be hidden, sensors may produce inaccurate readings, and the intentions of moving people may remain unknown.

B. Determinism: Stochastic

The real-world environment of the delivery robot is stochastic because the outcome of an action may be affected by uncertain factors, even when the same action is performed under similar conditions.

- **Predictable outcome:** Applying the brakes on a level, unobstructed surface will normally reduce the robot's speed.
- **Uncertain outcome:** A pedestrian may suddenly enter the robot's path, forcing it to stop or change direction.

Other sources of uncertainty include wheel slippage, sensor noise, changes in floor conditions, and unpredictable movements of other agents.

C. Simplified Simulator

In a simplified simulator, the delivery robot may operate in a fully observable and deterministic environment if the complete map and all relevant object positions are available and the simulator uses fixed rules for movement and obstacle behaviour.

However, these assumptions do not fully represent real-world conditions, where some variables remain hidden and action outcomes may be uncertain.

3 LLM-Based Student-Support Agent

A. Observability: Partially Observable

The student-support agent operates in a partially observable environment because it cannot automatically access all information about a student's circumstances, intentions, or academic records.

- **Observable variable:** The text of a student's question submitted through the chatbot.
- **Hidden variable:** A special deadline extension granted to the student but not included in the agent's available information.

The agent can use the conversation and approved knowledge sources, but missing context may prevent it from understanding the student's complete situation.

B. Determinism: Stochastic

A typical LLM-based student-support agent is treated as stochastic because its responses may vary depending on the model's generation settings, the wording of the question, and the available context. Retrieved information and external services may also change.

- **Predictable outcome:** A fixed rule maps a specific FAQ identifier to its corresponding approved university webpage.
- **Uncertain outcome:** An ambiguous question may be interpreted incorrectly, or the agent may generate a fluent but inaccurate response.

The use of an LLM does not guarantee that every answer will be correct or consistent.

C. Simplified Simulator

A scripted student-support chatbot that uses fixed input-output rules can be deterministic because the same input produces the same predefined response. It may also be treated as fully observable with respect to its limited internal state if all relevant variables are explicitly available.

However, this classification applies only to the simplified model. A real LLM-based agent may lack important student context, and its generated responses may contain errors or uncertainty.

4 Comparison Table

AI System	Observability	Justification	Determinism	Justification
Delivery Robot	Partially observable	Sensors cannot detect every hidden obstacle or environmental condition.	Stochastic	Pedestrian movements, wheel slippage, and sensor noise create uncertainty.
LLM-Based Student-Support Agent	Partially observable	The agent cannot automatically access all student circumstances or records.	Stochastic	Generated responses and interpretations may vary, and retrieved information may change.

5 Conclusion

Both the delivery robot and the LLM-based student-support agent are classified as partially observable and stochastic in their realistic operating environments. The delivery robot faces uncertainty due to physical conditions and moving objects, whereas the student-support agent faces uncertainty due to incomplete context, ambiguous questions, and variable language-model outputs.

A simplified simulator may make either environment fully observable or deterministic by removing hidden variables and unpredictable outcomes. Therefore, the classification depends on the assumptions and level of detail used to model the environment.